

# FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Annual Examination 2026

## Biology — Class 9 (Science Group)

**Syllabus: Chapter 1 — The Science of Biology | Chapter 2 — Biodiversity | Chapter 3 — The Cell | Chapter 4 — Cell Cycle | Chapter 5 — Tissues, Organs and Organ Systems | Chapter 6 — Molecular Biology | Chapter 7 — Metabolism (Sections 7.1 to 7.6 only)**

**Total Marks: 36**

**Time: 1 Hour 45 Minutes**

**Paper: Exam-16 (Version 1)**

1. Attempt ALL questions. There is no choice or OR alternative anywhere in this paper.
2. Section A (20 marks): 20 MCQs. Fill the correct bubble. Suggested time: 25 minutes.
3. Write complete working/reasons where a question asks for them. Read every question carefully.
4. Use black or blue ink only. Pencil is allowed only for the diagram in Section C.

### SECTION A — Multiple Choice Questions (20 × 1 = 20 Marks)

| # | Q. Choose the best possible option.                                                                                                                                                                                                                                                                                  | A                      | B              | C                      | D                   | Answer                                                                                                      |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------|------------------------|---------------------|-------------------------------------------------------------------------------------------------------------|
| 1 | In the malaria experiment described in the textbook, the blood of 100 healthy persons was examined under a microscope and <i>Plasmodium</i> was found in 7 of them. What fraction of the 100 healthy persons had NO trace of <i>Plasmodium</i> in their blood?                                                       | 7/100                  | 93/100         | 100/107                | 50/100              | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 2 | A statement that describes an observable occurrence in nature which appears to always be true, but does not explain WHY it happens (only THAT it happens), is called a:                                                                                                                                              | hypothesis             | deduction      | scientific theory      | scientific law      | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 3 | Bioinformatics, one of the emerging careers of biology mentioned in the textbook, is a combination of biology and:                                                                                                                                                                                                   | information technology | chemistry      | agriculture            | medicine            | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 4 | Binomial nomenclature requires a scientific name to be written with the genus name capitalized, the species name in lower case, and the whole name italicized (or underlined by hand). Which of the following correctly follows these rules for the lion, whose genus is <i>Panthera</i> and species is <i>leo</i> ? | PANTHERA LEO           | panthera leo   | <i>Panthera leo</i>    | <i>Panthera Leo</i> | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 5 | Multicellular, heterotrophic eukaryotes that lack a cell wall and chlorophyll, and include invertebrates like insects as well as vertebrates like fish and frogs, belong to kingdom:                                                                                                                                 | Animalia               | Fungi          | Plantae                | Protista            | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 6 | A network of protein fibres inside a eukaryotic cell, made up of microtubules, microfilaments and intermediate filaments, gives the cell its shape and helps it move. Which of the following is NOT one of these three components, according to the textbook?                                                        | Microtubules           | Microfilaments | Intermediate filaments | Ribosomes           | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 7 | Which layer of the plant cell wall is formed only after the cell has stopped growing, getting its extra thickness and rigidity from a substance called lignin?                                                                                                                                                       | Middle lamella         | Secondary wall | Primary wall           | Plasma membrane     | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 8 | During cytokinesis in an ANIMAL cell (but not in a plant cell), the plasma membrane pinches inward at the middle of the cell, eventually splitting it into two daughter cells. According to the textbook, this pinching-in structure is called a:                                                                    | middle lamella         | phragmoplast   | cleavage furrow        | cell plate          | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |

| #  | Q. Choose the best possible option.                                                                                                                                                                                                                                                              | A                                          | B                                    | C                                | D                                                                           | Answer                                                                                                      |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------|----------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 9  | A somatic (diploid) cell of a plant has 14 chromosomes ( $2n = 14$ ). This cell undergoes meiosis to form gametes. How many chromosomes will be present in each cell produced at the end of MEIOSIS I?                                                                                           | 7                                          | 14                                   | 28                               | 4                                                                           | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 10 | A somatic cell with $2n = 10$ chromosomes replicates its DNA during the S phase of interphase. At late prophase of MITOSIS (after S phase), how many chromosomes and how many chromatids per chromosome will this cell have?                                                                     | 10 chromosomes, 1 chromatid each           | 20 chromosomes, 1 chromatid each     | 5 chromosomes, 2 chromatids each | 10 chromosomes, 2 chromatids each                                           | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 11 | Besides providing support, protection and (with the help of muscles) movement, the skeletal system also helps maintain homeostasis by regulating the level of which mineral in the blood, according to Table 5.2 of the textbook?                                                                | Sodium                                     | Potassium                            | Calcium                          | Iron                                                                        | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 12 | According to Table 5.2 of the textbook, leukocytes, tonsils, adenoids, thymus and spleen are organs/structures of which organ system, whose homeostatic role is to remove disease-causing microbes from the body?                                                                                | Immune system                              | Lymphatic system                     | Digestive system                 | Endocrine system                                                            | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 13 | According to Table 5.2 of the textbook, the muscular system consists of skeletal, cardiac and smooth muscles, and along with the integumentary system, it helps maintain homeostasis by:                                                                                                         | filtering nitrogenous waste from the blood | producing antibodies                 | transporting oxygen              | regulating body temperature                                                 | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 14 | Which of the following nitrogen bases has a DOUBLE-ring structure, according to the textbook?                                                                                                                                                                                                    | Cytosine                                   | Guanine                              | Thymine                          | Uracil                                                                      | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 15 | Lipids cannot dissolve in water directly and must be transported in blood bound to carrier proteins. This general property, which applies to ALL lipids (unlike being "saturated" or "unsaturated," which applies only to some fatty acids based on their carbon-carbon bonds), is called being: | hydrophobic                                | hydrophilic                          | saturated                        | unsaturated                                                                 | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 16 | According to the textbook, approximately how many different types of amino acids are commonly found in proteins?                                                                                                                                                                                 | 4 to 6                                     | 10 to 15                             | 20 to 25                         | 30 to 35                                                                    | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 17 | The covalent bonds between the phosphate groups of an ATP molecule, which store energy that can be released for cellular work, are called:                                                                                                                                                       | Peptide bonds                              | High-energy bonds                    | Glycosidic bonds                 | Hydrogen bonds                                                              | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 18 | After an enzyme has converted its substrate into product, what happens to the enzyme molecule itself, according to the textbook?                                                                                                                                                                 | It is broken down along with the substrate | It permanently loses its active site | It becomes part of the product   | It regains its original shape and can be reused for the same reaction again | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 19 | When several enzymes work together in a specific sequence, with each one acting on the product of the previous reaction until an initial substrate is converted into a final product, this series of connected reactions is called a:                                                            | feedback loop                              | coenzyme cycle                       | metabolic pathway                | catalytic chain                                                             | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |
| 20 | The digestion of food in the stomach and intestine, where large, complex food molecules are broken down into smaller, simpler molecules, is an example of:                                                                                                                                       | anabolism                                  | catabolism                           | metabolism only                  | enzymatic hydrolysis                                                        | <input type="checkbox"/> A <input type="checkbox"/> B <input type="checkbox"/> C <input type="checkbox"/> D |

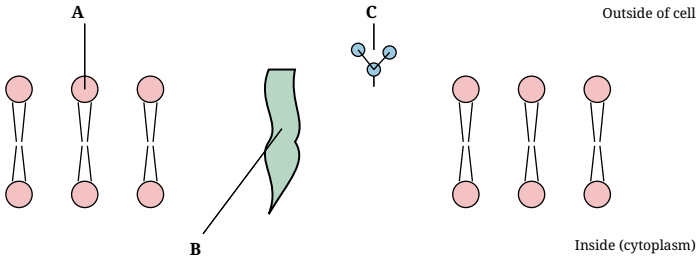
**SECTION B — Short Questions ( $2 \times 4 = 8$  Marks)**

Attempt ALL questions. There is no choice or OR alternative anywhere in this section.

| Part  | Question                                                                                                                                                                                                                                                     | Marks |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2(i)  | A polypeptide chain does not remain in its original straight-chain form after being made by a ribosome. Trace its folding journey through all FOUR structural levels of a protein, briefly stating what happens at each level, as explained in the textbook. | 4     |
| 2(ii) | Why is it impossible to eradicate malaria? Give at least four reasons based on the biology of <i>Plasmodium</i> and its mosquito vector, as discussed in the textbook.                                                                                       | 4     |

**SECTION C — Long Question ( $1 \times 8 = 8$  Marks)**

Attempt this question. There is no choice or OR alternative.

| Part | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 3    | <p>The diagram below shows the fluid mosaic model of the cell membrane, as given in the textbook (Fig. 3.4). Three parts have been left unlabelled.</p>  <p>(a) Identify the parts labelled A, B and C in the diagram above. [3]<br/>(b) State the approximate chemical composition of the cell membrane (percentage of proteins and of phospholipids), as given in the textbook. [2]<br/>(c) In 2–3 sentences, explain what the fluid mosaic model postulates about the arrangement of proteins in the cell membrane. [3]</p> | 8     |

Student's Signature: \_\_\_\_\_

Invigilator: \_\_\_\_\_

Date: \_\_\_\_\_

Chapters Covered: Chapter 1 – The Science of Biology, Chapter 2 – Biodiversity, Chapter 3 – The Cell, Chapter 4 – Cell Cycle, Chapter 5 – Tissues, Organs and Organ Systems, Chapter 6 – Molecular Biology, Chapter 7 – Metabolism (Sections 7.1 to 7.6 only)

— End of Question Paper —